

TUSHAR BHARADWAJ

Spatial Data Scientist | Remote Sensing & Geospatial ML

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PROFESSIONAL SUMMARY

Remote Sensing and Spatial Data Scientist with 6.5+ years specializing in geospatial ML, satellite image analysis, and scalable cloud-based pipelines. Proven track record designing and deploying CNN and deep learning models for forest monitoring, LULC classification, biomass estimation, and time-series change detection across millions of hectares using Sentinel-1/2, Landsat 8/9, and GEDI data. Experienced in leading multi-disciplinary teams of 5–6, delivering 6+ end-to-end geospatial ML projects, and conducting capacity-building workshops on GEE and computer vision for remote sensing. Passionate about applying AI-driven spatial analysis to ecosystem health, regreening, and climate resilience challenges.

CORE COMPETENCIES

Machine Learning & Deep Learning

CNNs / Image Classification	LSTMs	TensorFlow	Feature Engineering	Predictive Modelling	Model Validation
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Remote Sensing & Geospatial

Sentinel-1 SAR Processing	Sentinel-2 / Landsat 8-9	Time-Series Analysis	LULC Change Detection	Biomass Estimation	GEDI / LiDAR
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Cloud, Tools & MLOps

Google Earth Engine (GEE)	GCP / BigQuery	Python (Advanced)	Docker	GDAL / Rasterio	QGIS / ArcGIS
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PROFESSIONAL EXPERIENCE

Remote Sensing & GIS Lead | Thryve Earth

Mar 2024 – Present

- Architect and deploy ML/CNN pipelines for forest cover change detection, degradation mapping, and ARR/REDD+ prospecting, achieving 85–90% model accuracy across multi-million hectare project areas.
- Solely designed and delivered a full-stack geospatial decision-support platform (React, Python, GCP, Heroku) serving GIS specialists, NbS specialists, farmers, implementation partners, and investors across 5 districts in Karnataka and 3 in Indonesia — integrating eligibility assessment, site suitability mapping, carbon modeling, and farmer onboarding into a single end-to-end workflow.
- Built GEE-based eligibility and suitability dashboards for carbon project site selection, enabling rapid spatial screening of candidate landscapes for ARR/REDD+ registration.
- Developed automated carbon modeling and MRV dashboard integrating satellite-derived biomass, LULC, and field data to support VCS & Gold Standard project reporting.
- Lead Sentinel-1 SAR and Sentinel-2/Landsat 8/9 data fusion workflows for biomass estimation and carbon accounting; develop CNN-based LULC classification models for high-resolution land cover mapping and project boundary delineation.
- Design and maintain high-throughput satellite data pipelines on GCP (Cloud Storage, BigQuery) using Python and Docker, enabling scalable reproducible processing for large-scale land monitoring.
- Supervise and mentor a team of 5–6 junior spatial data scientists and developers on geospatial ML workflows, code quality, and cloud pipeline best practices.
- Contribute to proposal development and PDD writing for multiple carbon project registrations, EUDR compliance funding, and forest monitoring grants (IKI large grants).

Spatial Data Scientist & Research Fellow | Plant-for-the-Planet (NGO) · IIT Roorkee

Apr 2022 – Mar 2024

Applied research role combining NGO forest intelligence delivery (10 hrs/week) with academic ML & computer vision research — both focused on satellite-based ecosystem monitoring.

- Led end-to-end development of the TFFF Watch Tool — a production geospatial ML platform monitoring millions of hectares of tropical forest using Sentinel-2, Landsat 8/9, and GEE; built as a standalone web application (React, Python, GCP) serving NGO field staff and stakeholders.
- Built an LSTM-based vegetation trend forecasting model on Sentinel-2 vegetation index time series for anomaly detection and predictive monitoring of forest health dynamics.
- Designed EUDR compliance system (Tracer) integrating multi-source land-use, deforestation, and commodity traceability datasets; deployed as a GCP-hosted web application for implementation partners and investors.

- Contributed to project proposal development for TFFF compliance funding and wrote funding proposals for forest monitoring initiatives including IKI large grant applications.
- Built CNN-based computer vision pipelines for urban forestry using multispectral and LiDAR data; developed tree canopy extraction and biomass estimation models (IIT Roorkee).
- Delivered a capacity-building workshop on GEE and computer vision for remote sensing, training 25 students; created interactive GEE dashboards using JavaScript for real-time stakeholder engagement.

Assistant Professor — GIS & Remote Sensing | Govt. College of Technology & Engineering Sep 2019 – Feb 2022

- Taught ML-driven geospatial analytics, Python-based spatial data analysis, and GIS; supervised applied remote sensing and ML research projects.
- Mentored students on real-world geospatial problem-solving; contributed to curriculum modernization integrating cloud-based GIS tools.

KEY PROJECT HIGHLIGHTS

TFFF Watch Tool — Production ML platform for tropical forest monitoring across millions of hectares (Sentinel-2, GEE, GCP)

EUDR Compliance Tool (Tracer) — Geospatial ML web application integrating deforestation, land-use and supply-chain datasets (GCP, Python)

Global Biomass Estimation Model — Multi-source pipeline fusing GEDI, SAR, and optical data with ML for large-scale above-ground biomass mapping

Carbon Prospecting Platform — ML-based ARR/REDD+ suitability analysis tool for carbon project prospecting and MRV automation

EDUCATION

PG Diploma in Data Science | IIIT Bangalore (GPA: 3.12/4.0)

2020 – 2022

GIS & Remote Sensing Certification | IIT Roorkee (9.5/10)

M.Tech – Geotechnical Engineering | NIT Hamirpur (CGPA: 7.6/10)

B.Tech – Civil Engineering | Rajasthan Technical University

PUBLICATIONS

Bharadwaj, T., Kumar, A., & Khare, S. (2023). *Smart agriculture platforms: Case studies*. In Technological Prospects and Social Applications of Society 5.0 (pp. 169–176). CRC Press.

Khare, S., Bharadwaj, T., et al. (2023). *Machine learning and satellite-based multi-scale segregation of coniferous and deciduous trees*. GEO BON Global Conference: Monitoring Biodiversity for Action.

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent) | **Availability:** Open to remote work